HANGCHA
Making Handling Easie

User Manual



Lithium battery charger

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Chapter 1 General

1.1 Instruction

The safety precautions, product features, technical specifications, operating procedures, installation and debugging, troubleshooting, equipment maintenance, and other contents of the user manual for the HCD Lithium Battery Smart Charger are elaborated and explained. Users are advised to carefully read this manual after unboxing and keep it properly for future reference.

Our company reserves the right to modify the manual without prior notice, and we reserve the right not to notify separately to the public.

1.2 Safety Warning



1.3 Application Regulation

★When using and connecting the lithium battery charger, strict compliance with the relevant regulations for EV charging is required.

★The lithium battery charger has been developed, produced, inspected, and filed according to relevant safety standards. If all instructions and safety technical instructions are followed, under normal circumstances, it will not cause property damage or endanger personal safety.

★The instructions contained in this manual must be strictly followed; otherwise, safety hazards may arise or safety devices may fail. Although this manual explains relevant safety instructions, it is still necessary to comply with safety regulations and accident prevention regulations in the area of product use.

1.4 Target User

This manual is applicable to the following groups of people:

★Charging equipment training personnel;

★Charging operators;

★Technical service personnel.

Chapter 2 Safety Precautions

2.1 Safety Statement

Failure to comply with the safety precautions in this section while using this equipment may result in life-threatening situations, injuries, and equipment damage! Hangzhou Hangcha Yunrui Technology Co., Ltd does not assume any liability for claims resulting from such incidents.

2.2 Safety Tips

To ensure personal and equipment safety, personnel operating this equipment must adhere to the following principles and regulations:

★Before powering on the equipment, ensure that the equipment is properly grounded.

★Before powering on the equipment, ensure that the input voltage, frequency, circuit breakers or fuses of the device, and other conditions meet the specifications.

★All tools should be properly insulated to prevent exposed metal parts from coming into contact with metal frames, causing short circuits.

★Ensure stable operation of the equipment. The equipment should be operated in a clean, well-ventilated environment, and the operating environment must not contain volatile or flammable gases.

2.3 Safety Regulation

In this manual, different symbols are used to indicate potential dangers at various locations. The meanings of these symbols are as follows:

No.	Safety Symbol		Definition
1		Danger	May cause death or serious injury if appropriate precautions are not taken
2		Warning	May cause property damage or minor injury if appropriate precautions are not taken
3		Caution	May cause property damage if appropriate precautions are not taken.

1. Danger

★Electrical hazard, fire hazard! Do not use damaged, worn-out, or dirty charging guns.

★Do not place flammable, explosive, or combustible materials, chemicals, combustible vapors, or other dangerous substances near the charging equipment.

★Power must be disconnected during equipment installation and replacement to prevent electric shock.

★Do not use any extension cables to connect the charging equipment to electric forklift trucks. The charging equipment is only for connecting to electric forklift trucks, not for connecting to other loads (such as power tools).

★Avoid mechanical damage to the charging cable (bending, squeezing, or rolling), and do not allow the cable surface to contact heat sources, dirt, or water sources.

★All equipment cables must be securely connected, well-insulated, and of appropriate size. Loose, damaged, or undersized cables may cause circuit failures, leading to irreparable damage to the equipment, or even causing injuries, fatalities, or fires.

★Even if the equipment has not sustained visible damage, do not install it directly if it has been dropped. Some internal damage may lead to system disturbances or faults such as circuit short circuits. It must be inspected by professionals to determine if it is safe to install.

★Never rinse the equipment with water, as water ingress can cause circuit short circuits.

2. Warning

★Any faults that may affect its safety performance must be immediately rectified by professionals before restarting.

★When plugging or unplugging the charging gun, grip the handle tightly and do not pull the cable.

★Do not allow heavy objects to press on the cable or step on the charging cable deliberately.

★Keep the charging gun clean and dry. If it is dirty, wipe it with a clean dry cloth. Do not touch the charging gun core directly with your hands.

★The installation of the product should be carried out by professional technical personnel from the company (or authorized professional institutions). Unauthorized installation or modification may cause safety accidents or equipment damage. The company shall not be liable for any losses caused thereby.

3. Caution

★If any abnormal situation occurs during use, press the emergency stop button immediately to cut off all input and output power.

★Use caution in rainy or thunderstorm weather.

★The normal operation of the equipment depends on the reasonable setting of some parameters, which are generally set after on-site installation and commissioning. If not necessary, it is not recommended for users to reset the parameters by themselves.

★Charging can only be performed after the vehicle is turned off, and the vehicle can only be started after charging is complete. During charging, the electric forklift trucksis prohibited from driving.

★Regularly check whether the charging gun and charging cable are damaged, and inspect whether the equipment casing is damaged.

★When using the charging equipment to charge electric forklift trucks, please read the relevant prompts and user manuals of the vehicles carefully.

★The warnings and operation signs inside the equipment body and cabinet play an important role in the safe operation of the equipment. It is strictly prohibited to tear or damage them. If they are damaged or blurred, they should be replaced promptly.

Chapter 3 Product Introduction

3.1 Reference Standard

(1) GB/T 18487.2-2017 Electric vehicle conductive charging connection system Part 2: Non-vehicle-mounted conductive supply equipment electromagnetic compatibility requirements

(2) GB/T 34657.1-2017 Electric vehicle conductive charging interoperability test specification Part 1: Supply equipment

(3) GB/T 34658-2017 Communication protocol consistency test between non-vehicle-mounted conductive chargers and battery management systems for electric vehicles

(4) GB/T 27930-2023 Communication protocol between non-vehicle-mounted conductive chargers and battery management systems for electric vehicles

(5) GB/T 18487.1-2015 Electric vehicle conductive charging system Part 1: General requirements

(6) GB/T 20234.1-2015 Electric vehicle conductive charging connection device Part 1: General requirements

(7) GB/T 20234.3-2015 Electric vehicle conductive charging connection device Part 3: DC charging interface

(8) GB/T 17626.2-2018 Electromagnetic compatibility Test and measurement techniques Electrostatic discharge immunity test

(9) NB/T 33001-2018 Technical conditions for non-vehicle-mounted conductive chargers for electric vehicles

(10) NB/T 33008.1-2018 Inspection and test specifications for electric vehicle charging equipment Part 1: Non-vehicle-mounted chargers

(11) Q/TT001-2019 Non-vehicle-mounted DC chargers for electric vehicles

3.2 Working Condition

★Altitude does not exceed 2000m;

★Operating environment temperature: -20°C~ +50°C;

★The maximum relative humidity of the surrounding air does not exceed 95% (when the ambient air temperature is 25± 5°C);

★The installation base should be stable, without severe vibration or impact, with a vertical inclination not exceeding 5%;

★The operating location should be free from conductive or explosive dust, corrosive gases, or vapors that damage insulation;

★When operating outdoors, a canopy should be installed above the equipment to prevent rain.

3.3 Product Overview

The HCD lithium-ion intelligent charger is a new type of intelligent charger developed, designed, and produced by Hangzhou Hangcha Yunrui Technology Co., Ltd. based on years of on-site application and power electronics technology. The product features high input power factor, low input current harmonic, high efficiency, good EMC performance, simple process, high standardization, high reliability, aesthetic appearance, and simple use, operation, and maintenance. It is far superior to similar products in the industry and can be widely used in transportation, industry, logistics, and other industries.

3.4 Product Features

★Adopting fully digital control technology

Advanced digital circuit control system design replaces traditional analog circuits. High-speed microcontrollers sample and self-test all functional circuits throughout the process, accurately judge the real-time working status of each functional circuit, and conduct fault analysis to ensure the system operates reliably and stably.

★Intelligent detection and monitoring functions

The microprocessor of the digital circuit control system continuously monitors the output power status, circuit working status, and battery operating status in real-time. When a fault occurs, the system will promptly alarm and initiate protection functions synchronously.

★Electrical isolation design

To improve the reliability of the system operation, an electrical isolation circuit architecture is adopted for both input and output.

★Input rectification part

Advanced and mature circuit design is used to improve the input power factor and reduce input harmonic current; the average input current tracks the reference current without being constrained by the load current, so even if the load current has significant harmonics, the input current will not undergo severe distortion.

★Comprehensive protection functions

It has various comprehensive protection functions and alarm mechanisms, including input under/over-voltage protection, input surge protection, output overload and short-circuit protection, over-temperature protection, etc.

★Forced air cooling design

Efficient cooling air duct design. Multiple high-power fans are used for effective forced cooling.

★Complete interface configuration

Input/output power interfaces, CAN communication interfaces, and reserved interfaces, etc.

★Modular production process and assembly process design

Reasonable internal layout and simple wiring avoid complex production processes.

★Intuitive display of working parameters and working status, simple interface operation.

3.5 Working Principle

AC power is rectified by a bridge stack and applied to the inverter main circuit composed of IGBT modules and fast recovery diodes to form a high-frequency pulsating DC. The high-frequency pulsating DC is coupled and isolated through a high-frequency transformer and then rectified by fast recovery diodes to output energy to the load. The system adopts PWM pulse width modulation, adjusts the width of the pulse according to the detected output current and voltage signals, changes the duty cycle to stabilize the output. At the same time, current and voltage can be limited according to the working conditions. The entire charging process is intelligently controlled by DSP to ensure the high reliability of the charger. In the design, overvoltage

protection, short-circuit protection, overheat protection, delayed start protection, and other protection measures are provided to ensure the safe operation of the system.

3.6 Technical Spec of the Charger

Item	Parameter Index				
Model	HCD（CE）-9680	HCD（CE）-96100	HCD（CE）-96200	HCD（CE）-48100	HCD（CE）-48200
Input Technical Parameters					
Input Voltage	AC400V±15%				
Input Current	< 13A	< 16A	< 32A	< 10A	< 19A
Frequency	50Hz~60Hz				
Power Factor	≥0.99				
Current Harmonic	≤5%				
Output Technical Parameters					
Maximum Voltage	DC110V	DC110V	DC110V	DC60V	DC60V
Maximum Current	80A	100A	200A	100A	200A
Maximum Power	8KW	10KW	20KW	6KW	12KW
Output Voltage Range	35~110VDC	35~110VDC	35~110VDC	35-60VDC	35-60VDC
Adjustable Current Range	3A-80A	3A-100A	3A-200A	3A-100A	3A-200A
Efficiency	>93%				
Voltage Ripple	<3%				
Current Ripple	<3%				
Voltage Regulation Accuracy	5‰				
Current Regulation Accuracy	1%				
Charging Curve	BMS designated				
Operating Environment Conditions					
Operating Temperature	Normal operation-20~50℃; Output reduction at 50~65℃				

Item	Parameter Index
Storage Temperature	-40~75℃
Relative Humidity	5~95%
Altitude	≤2000 full load output
Protection Functions	
Input Protection	Input under/over-voltage
Output Protection	Output over-voltage,output short circuit, battery reverse connection
Over-temperature	Protection Shutdown if internal module over-temperature
Abnormal Protection	Shutdown if power device is abnormal
Emergency Stop	Protection Force shutdown through emergency stop button when uncontrollable abnormality occurs
Interface	
CAN Interface	1 channel
485 Interface	Touchscreen model supports 1 channel
Human-machine Interaction	
Display mode	touchscreen
Appearance and Protection	
Cabinet Size Dimensions	Outer dimensions (main unit)：500mm (length) ×300mm (width) ×600mm (height)
Protection degree	IP20
Others	
Cooling method	Forced air-cooling

Chapter 4 Installation & Commissioning

4.1 Installation Precautions

The working voltage and current of certain components inside this power system are high. To ensure personal safety, the following regulations should always be followed:

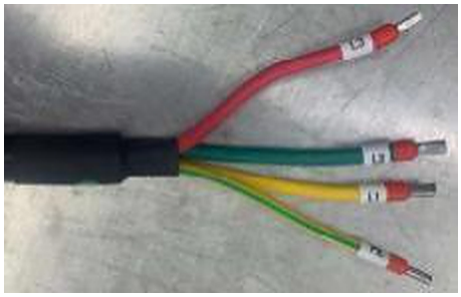
1.Only personnel who have received training on the power system and have a comprehensive understanding of the system are allowed to install this power system.During the installation

process, they must always adhere to the safety regulations listed above as well as local safety regulations.

2.When operating inside the power system cabinet, ensure that the power system is not energized. The mains power input and battery power of the power system must be disconnected.

3.Place the power equipment in a locked room, lock the cabinet, and have the person responsible for the power equipment keep the key.

4.2 Input Wiring Instructions



Without Plug



Yeeda Plug

★ Wiring Definition:

The input line is either three-phase four-wire or three-phase five-wire. The interface for three-phase four-wire is labeled as A, B, C, PE; the interface for three-phase five-wire is labeled as A, B, C, N, PE.

Explanation: ABC are the three phase lines, N is the neutral line, and PE is the earth line

⚠ Warning： PE grounding must be connected to the earth to ensure personal and equipment safety. The N line must not be connected incorrectly, as it may cause damage to internal components of the machine. The company will not be held responsible for accidents caused by incorrect AC input wiring by users.Thank you for your cooperation.

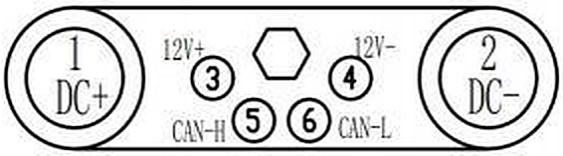
4.3 Output wiring description



REMA Plug

★ Wiring Definition:

This is the REMA output plug, its wiring definition is as shown in the diagram:





Output Woer Charging Gun

★ Wiring Definition:

According to GB/T 20234.3-2015, the wiring definition for the charging gun is as follows:

DC power supply lines: DC+, DC-

Equipment ground wire: PE

Charging communication lines: S+, S-

Charging connection confirmation lines: CC1, CC2

Low-voltage auxiliary power supply lines: A+, A-

4.4 Commissioning Instructions

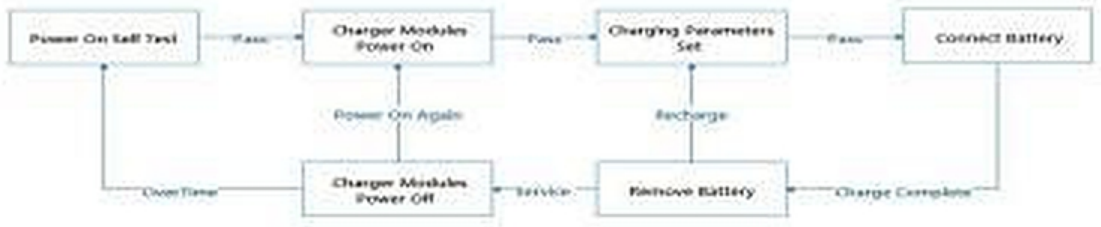
4.4.1 Precautions Before Powering On

★Before powering on the charger, check whether the battery box connections are correct to prevent short circuits or reverse polarity. This precaution helps avoid overloading and burning components or circuits in the charger.

★The charger can only be put into operation after confirming that there are no abnormal conditions or overheating warnings during startup.

★To ensure the safety of the charger and equipment operation, it is strictly prohibited to directly disconnect the battery switch when there is output current, except in emergency situations.

4.4.2 Operation flow chart



4.4.3 Specific Operations

(1) Power-On Inspection

Ensure that the AC input line and DC output positive and negative poles are correctly connected, with no short circuits in input or output; verify that input voltage and frequency are normal; always remain in standby power-on state.

(2) Power-On Startup

①After powering on, check if the AC fan direction is correct.

②The touchscreen should open normally, and communication with the charger should be normal.

(3) Battery Connection

①Battery voltage and current levels meet the charger requirements. Battery polarity is correct. The battery is in good condition without abnormalities.

(4) Power-Off Shutdown

①After the charging current and voltage drop to 0, remove the battery.

②Turn off the AC input circuit breaker

Chapter 5 Display Interface and Touch screen interface

1 Home Page Interface



★Click on the icon to enter the help menu.

★Click on the icon or to toggle the alarm sound. indicates the alarm sound is enabled, indicates the alarm sound is disabled.

★Control operations require password authorization. The default password is: 123456.

★Click the "Power on/off" icon to toggle the charging control between start and stop.

★Click "Reservation Settings" icon to switch between Normal Mode and Reservation Mode.

2 Help description interface



★Click on the icon to return to the main interface.

★Click on the “Data Export” icon to enter the data export interface.

★Data Export Settings Instructions:

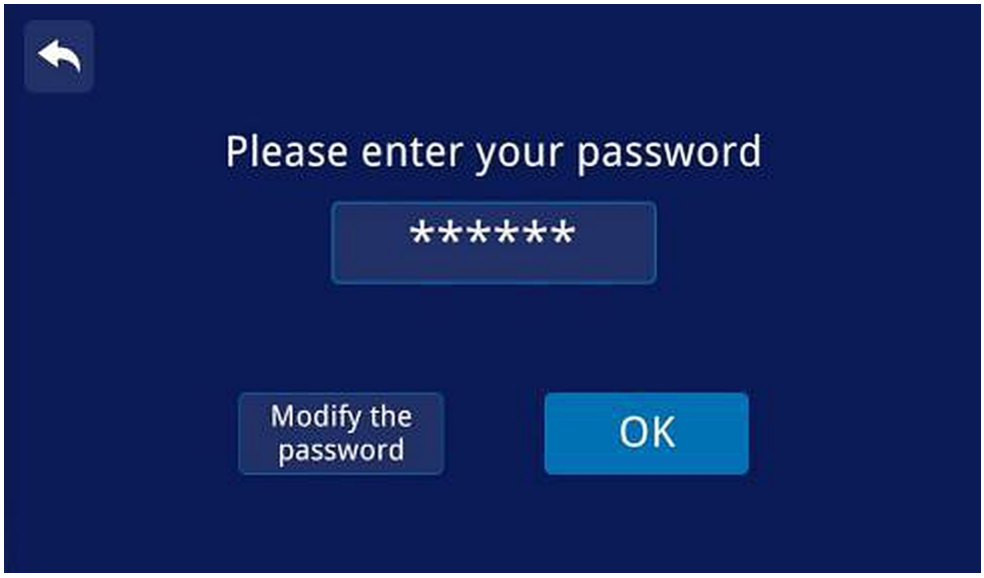


★Clicking the “Data Export” icon allows you to export charging records from the charger by inserting a USB drive or storage disk.

★Clicking the “Data Delete” icon enables you to delete the charging records of the charger.

★Click the “Factory Settings” icon to access the password input interface. Once the correct password is entered, you will be directed to the factory settings interface.

3 Password input interface



★Enter the correct password in the password input field. The default password is 888888.

★Click the icon to return to the Help Guide interface.

★Click the “Modify” icon to access the password modification interface.

★Click the “Confirm” icon. If the password is correct, you will enter theparameter settings interface. Otherwise, an error message will prompt you to re- enter the password.

4 Password change interface

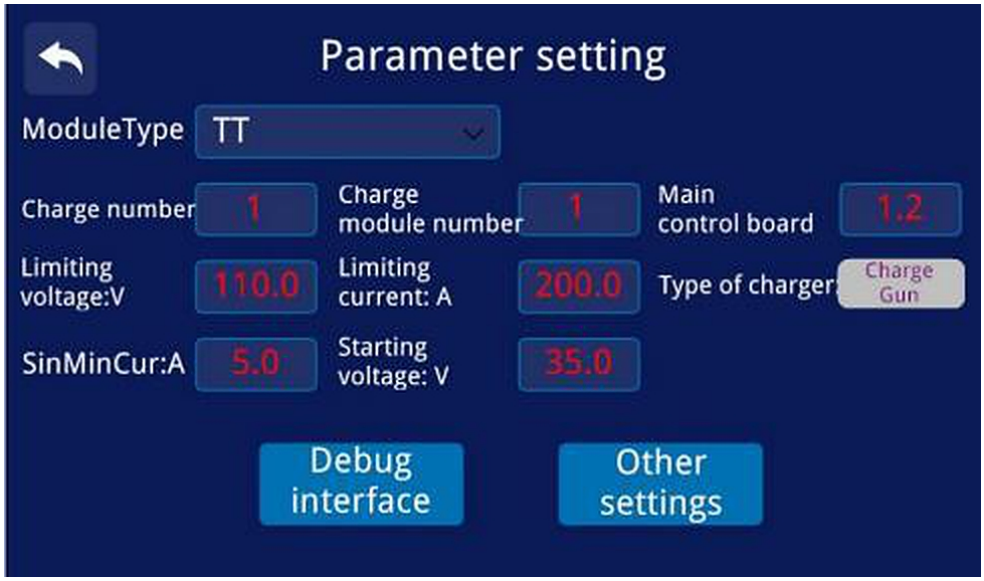


★Follow the prompts to enter your current password, then input the new password twice for confirmation.

★Click the “Confirm ” icon. If the current password is correct and both entries of the new password match, the password will be successfully updated. Otherwise, an error message will appear, and you'll need to try again.

★Click the icon to return to the password input screen.

5 Parameter setting interface

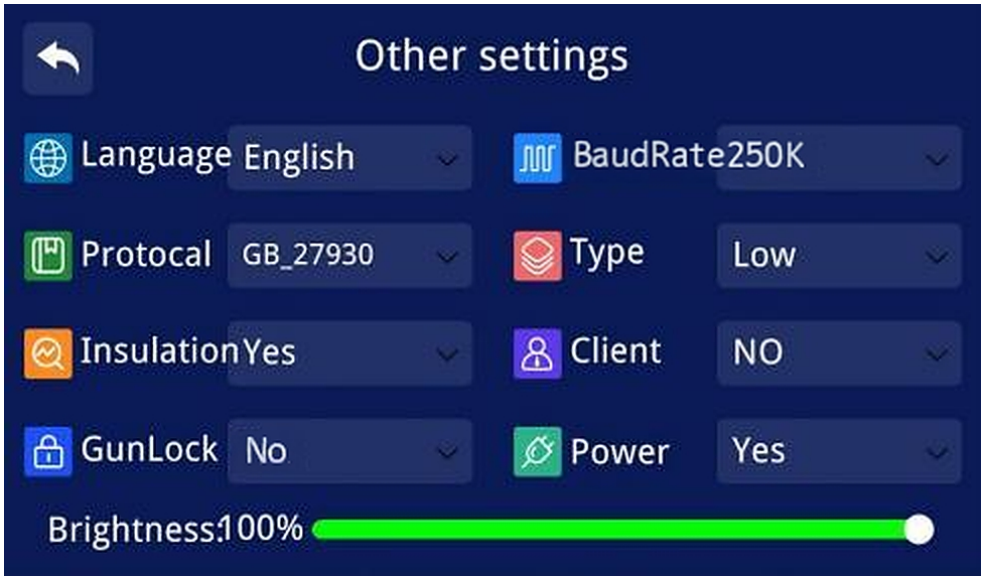


6 Other Setting

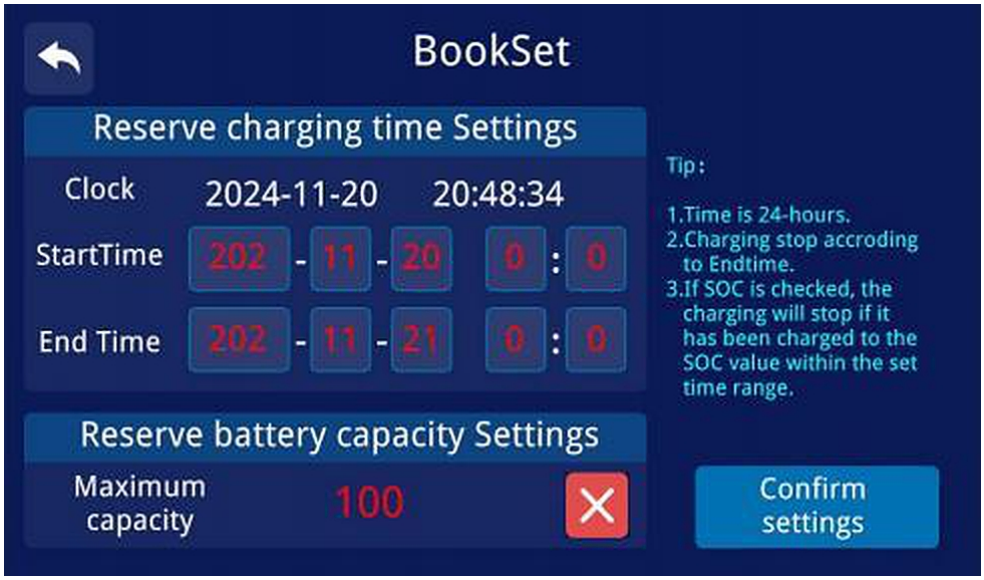
★Click the "Other Settings" icon to access the Other Settings interface.

★Select Language to switch between Chinese and English interfaces.

★Important: Do not modify other settings without assistance from the manufacturer. Unauthorized changes may prevent the charger from functioning properly.



7 Reservation charging



★On the main screen, click the "Scheduled Charging" icon and enter the password 123456 to access the scheduling settings.

★Specify the start and end times for the scheduled charging.

★Set the capacity limit and click the " ×" in the top-right corner to change it to " √".

★Once you click the Settings icon, the scheduling is saved, and the system automatically returns to the main screen.

8 Charging Display Interface



★Click the forklift icon on the main interface to enter the Charging Display screen.

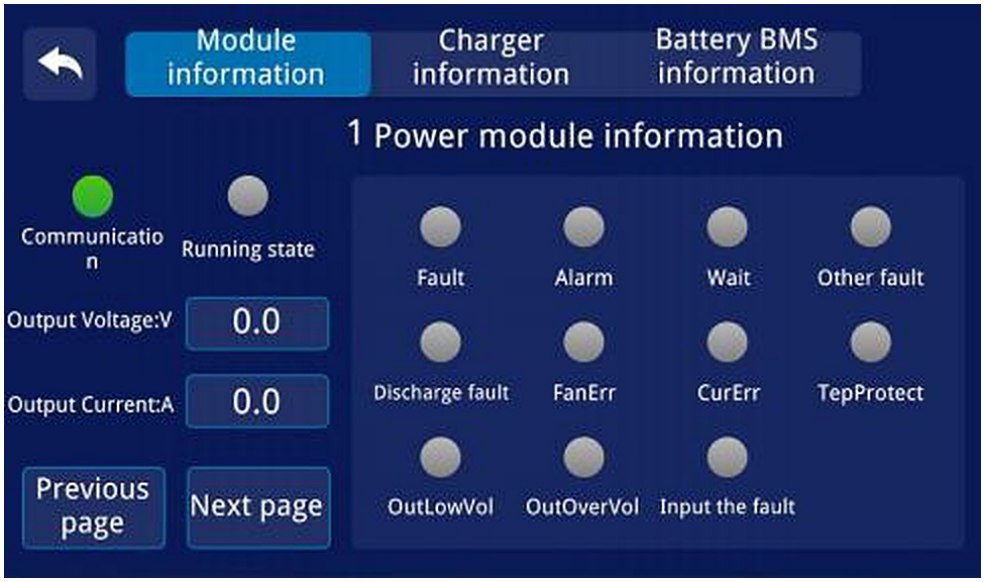
★This interface displays current charging parameters, including output voltage, output current, charging energy, charging duration, and the battery’ s current SOC (State of Charge).

★Click the “Stop Charging” icon to terminate the current charging session.

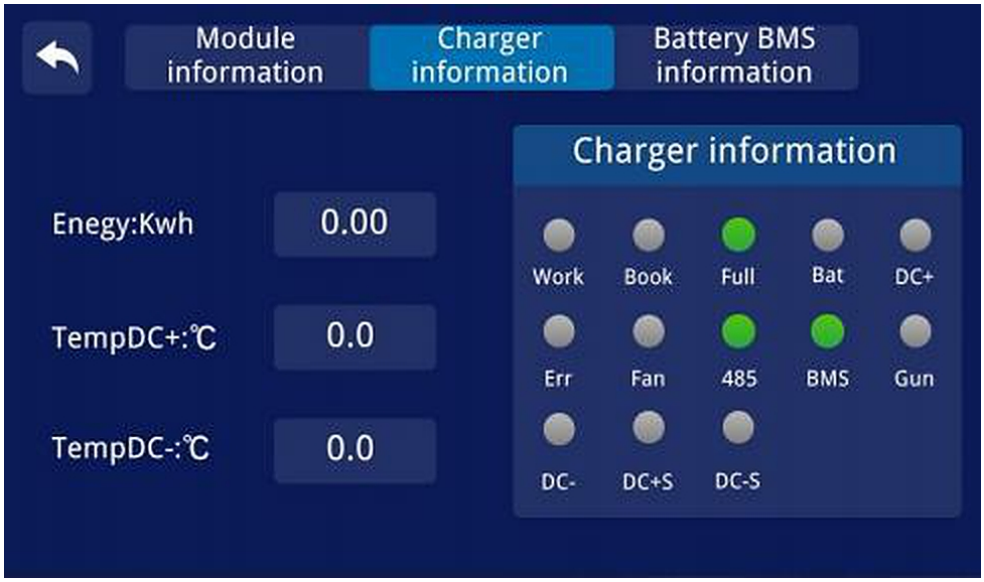
9 Charger Information Display

★Click the “Query” icon to access the Charger Information Status screen.

★As shown in the figure below, this screen provides details about charger module information, charger status, and battery BMS data.



★Module Information: Displays the main power output module details, current status, and fault information for each module. A red indicator light signals a module fault.



★Charger Information: Shows the charger's current operating status. A green light indicates normal operation, while a red light signals a corresponding fault.

★Battery BMS Information: Displays information received from the battery's BMS. Detailed content is available only for systems following the national standard protocol; custom protocols provide no detailed information.

★Use the “Previous Page” or “Next Page” buttons to view specific messages from the national standard protocol.

10 Detailed Explanation of the Charging Process



★ Charging Steps

1. Ensure the charger is properly connected to the power supply. Switch on the circuit breaker to power up the charger for normal operation.

2. Confirm that the DC charging connector is securely connected to the battery.

3. Charging will only begin if the “Remaining Capacity” is less than the “Allowed SOC Limit. ”

4. The Battery Indicator will illuminate green once the charger detects the battery voltage, and the BMS Indicator will also turn green.

5. The Working Indicator will light up green. When the “Output Voltage” is close to the “Battery Pack Voltage,” the charger’s output relay will engage, and charging will officially start. During this process, the “Output Current” and “Output Voltage ” will adjust based on the “Current Demand” and “Voltage Demand.”

6. Once the BMS completes charging, it will send a termination command, and the charger will stop charging.

7. Charging will stop under any of the following conditions:

- Charger faults such as Overheating or Abnormality (indicated by corresponding warning lights).

- Completion of charging as instructed by the BMS.

- "Charging Allowed" is set to prohibited.

- Manual intervention to stop the charging process.

★In Scheduled Mode, the charger will only begin operation at the preset scheduled time. ★

Chapter 6 Common Faults & Troubleshooting Methods

6.1 Troubleshooting Methods

When the charging machine is operating abnormally, you can find the charging status display area on the right side of the charging machine's touchscreen. By referring to the status indicators above, you can determine the cause of the malfunction and proceed with troubleshooting accordingly.

6.2 Solutions to common faults

6.2.1 BMS status indicator not lit - Causes:

1.Damage to the BMS of the vehicle's battery.

2.Faulty charging machine controller.

3.Damage to the internal 12V auxiliary power supply of the charging machine.

4.Poor connection of the charging gun.

Solutions:

1.Try charging another vehicle. If it works normally, the issue lies with the battery's BMS. If the problem persists, it indicates a fault with the charging machine's internal controller or the 12V auxiliary power supply.

2.Check if the charging gun is securely connected.

3. Regularly use an air gun to clean the interior of the charger.

6.2.2 CAN status indicator lit red - Causes:

1.Fault in the CAN communication port between the main control board and the module.

Solutions:

1)Check the connection line between the main control board and the module. If ineffective, proceed to step 2.

2)Replace the main control board. If ineffective, proceed to step 3.

3)Replace the charging module.

6.2.3 485 status indicator lit red - Causes:

1.Communication failure between the main control board and the touchscreen.

Solutions:

1.Check the connection line between the main control board and the touchscreen. If ineffective, proceed to step 2.

2.Replace the main control board. If ineffective, proceed to step 3.

3.Replace the touchscreen.

Chapter 7 After-sales Warranty

7.1 Warranty Conditions

★If users find product or accessory damage during unpacking due to transportation reasons.

★Quality problems occur while users fully comply with the installation, storage, and usage rules specified in this manual.

7.2 Warranty Period

The specific time is based on the contract signing date, starting from the date of product delivery.

7.3 Warranty Methods

★Within the warranty period, the manufacturer is responsible for free replacement or repair.

★If the warranty period is exceeded, users should negotiate with the manufacturer for paid replacement or repair.

7.4 Exemption of Liability

The following circumstances are not covered by the company's warranty:

★The entire machine or components have exceeded the warranty period.

★"Product damaged during transportation" due to user reasons.

★The equipment operates in extremely harsh environments beyond those described in this manual.

★Machine failures or damages caused by non-company service personnel repair, alteration, or disassembly.

★Any actions beyond the installation and usage scope specified in relevant national standards.

★Damage caused by human factors.

7.5 Non-responsible Replacement Terms

★Replacement must ensure that the product's appearance is intact, without scratches or deformations.

★Damage caused by irresistible fires, floods, lightning strikes, earthquakes, etc.

7.6 After-Sales Contact Information

After-Sales Manager:

Contact Person: 15382380131

Chapter 8 Packaging, Transportation, and Storage

8.1 Packaging

The charging equipment adopts a modular design, with the power module supporting quick disassembly and assembly (using a card slot fixing method). During transportation, it is not necessary to ensure the integrity of the DC charging machine. Depending on the equipment specifications, we will choose between packaging the equipment as a whole or separately packaging the charging cabinet and power modules. During transportation, ensure that internal

components of the equipment do not shift, become damaged, or dampened, and do not affect the installation and performance of the equipment.

8.2 Transportation

The product should not be subjected to severe vibration, impact, or inversion during transportation.

8.3 Storage

If the product is not used immediately after purchase and needs to be stored for short or long periods, it should be stored away from high temperatures, dust, and metallic powders. Ensure that the surrounding air does not contain acidic, alkaline, or other corrosive or explosive gases, and store it in a dry, well-ventilated indoor location.

8.3.1 Storage Environment

(1) Ambient temperature: -40°C to 75°C.

(2) Relative humidity: 5% to 95% (25°C).

(3) Altitude: ≤3000m.

(4) It should be stored in a clean, dry environment without corrosive gases and should not be mixed with chemicals.

(5) Moisture-proof pads such as wooden boards should be placed on the warehouse floor, and storage time should not be too long or piled up.

(6) The storage location should avoid drastic changes in the surrounding environment and direct sunlight exposure.

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If there are any changes to other content of this manual, we reserve the right to interpret it, and no further notice will be given. The final interpretation right belongs to Hangzhou Hangchayunrui Technology Co., Ltd.

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